

ATUL KUMAR SINGH

+91 9353955244 [✉ atul13062001@gmail.com](mailto:atul13062001@gmail.com) [in linkedin](#) [github](#)

Education

National Institute Of Technology Karnataka

Bachelor of Technology in Information Technology

Aug. 2019 – April 2023

7.77 CGPA

Technical Skills

Languages: Python, Java, HTML/CSS, JavaScript, SQL

Databases : MongoDB, Redis, MySQL, Snowflake

Frameworks: React.js, Next.js, Node.js, FastAPI

Tools/Technologies: Docker, Redis, Git, Langchain, Langgraph, Vector Databases. Machine Learning, Deep Learning

ML Architectures: NLP, Transformers, ANNs, LSTMs, LLMs, Retrieval Augmented Generation.

Experience

Carelon Global Solutions India

July 2023 - Present

Associate Software Engineer

Bangalore

- Engineered a configurable big data ML pipeline using PySpark & H2O PySparkling, processing **20M+ datapoints** while dynamically selecting from **18K+ features** via Random Forest-based importance, reducing processing time by 40%.
- Built a modular **ML workflow** allowing users to experiment with different **MLlib algorithms**, **configure hyperparameters**, and **automate model evaluation, selection, and AUC-ROC analysis**, ensuring reproducibility by storing datasets & artifacts of all steps within the pipeline..
- Revamped an automated transformation system to simplify complex medical claims, by integrating **AWS Textract** for OCR processing, integrated **AWS S3 storage** into the pipeline, reducing runtime by **15% for OCR processing**.
- Led enhancement to the pipeline by **caching OCR embeddings** to S3 storage, decreasing the runtime by 28%.
- Contributed to developing a **FastAPI-powered** chat system leveraging **LLM & RAG** to query over 35+ HR policy documents. Incorporated a **self-reflective adaptive RAG** using **LangGraph**, improving accuracy by allowing the model to validate and refine its own responses reducing hallucinations by 20%.

Carelon Global Solutions India

May 2022 - July 2022

Data Science Intern

Bangalore

- Performed **data cleaning and clustering** on patient discharge summaries, leveraging **NLP** to extract key insights for medical claims processing.
- Applied **machine learning** techniques like **sentence ranking**, **K-Means clustering**, and **Bio-BERT** for accurate summarization.
- Optimized performance by implementing **parallel processing**, reducing runtime by 20% at the line, paragraph, and page levels

Projects

CocoBOT | *Python, FastAPI, LLMs, React Native*

[Source Code](#)

- Orchestrated a **multi-agent chatbot system** (guard, details, order-taking, recommendation) to simulate human-like interaction in an e-commerce setting.
- Used open-source **Large Language Models with Retrieval-Augmented Generation (RAG)** to enable dynamic product queries and intelligent order handling.
- Implemented a recommendation engine using Apriori-based basket analysis to suggest frequently bought-together products.
- Developed a FastAPI backend with MongoDB and caching to ensure low-latency responses and optimal performance, designed a mobile UI with React Native for real-time user interaction.

Data Analysis Assistant | *Python, Streamlit, Langchain, Langgraph, Seaborn, Plotly*

[Source Code](#)

- Made an **agent-based assistant** enabling non-technical users to interact with and understand their datasets through **natural language queries**.
- Integrated Large Language Models (LLMs) with LangChain to interpret user questions and provide insightful data analysis responses and on the fly chart generations using plotly and seaborn.
- Designed a modular architecture for seamless CSV upload, efficient querying, and **real-time insights**.